

GraphTheorySymbol:

A set of $\text{\LaTeX}$ commands to have symbols designed for
Graph Theory

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Abstract

This documentation is for $\text{\LaTeX}$ package GraphTheorySymbol. It provides symbols used in graph theory. The adjacency and incidence symbols were created by the author of this package.

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1 Introduction

This documentation is for $\text{\LaTeX}$ package GraphTheorySymbol. It provides symbols used in graph theory. The adjacency and incidence symbols were created by the author of this package.

This package is licenced under Lesser GNU General Public License version 3 or later (LGPLv3+). It means you are free do anything with it, but if you do improve the code of the symbols that we provide, please do share these improvements in a similar way. There is no restriction on other graph theory symbols/code, even if it would be a good fit for inclusion in this package. There is also no need to share in a similar way what you build with this package. The text of the licenses is available at:

<https://www.gnu.org/licenses/>.

The git repository of this source code is also available at:

<https://github.com/LLyaudet/GraphTheorySymbol/>.

Clone it with this URL:

[git@github.com:LLyaudet/GraphTheorySymbol.git](https://github.com:LLyaudet/GraphTheorySymbol.git).

This package is available on CTAN:

<https://ctan.org/pkg/graph-theory-symbol>.

This is our first $\text{\LaTeX}$ package. It may be that our macros are quite slow and that a document making extensive use of these macros will be slow to compile. If you know how to produce better $\text{\LaTeX}$ code, please submit a pull request on GitHub.

2 Usage

2.1 Memo or for the impatient

Add in your preamble: `\usepackage{graph-theory-symbol}`.

Long name	Short name	Result	In context
<code>\GTSadjacency/</code>	<code>\GTSa/</code>	$\circ$	$u \circ v$
<code>\GTSadjacencyRelation/</code>	<code>\GTSaR/</code>	$\circ$	$u \circ v$
<code>\GTSunadjacency/</code>	<code>\GTSu/</code>	$\circ$	$u \circ v$
<code>\GTSunadjacencyRelation/</code>	<code>\GTSuR/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyUp/</code>	<code>\GTSdAU/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyUpRelation/</code>	<code>\GTSdAUR/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyUp/</code>	<code>\GTSnDAU/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyUpRelation/</code>	<code>\GTSnDAUR/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyDown/</code>	<code>\GTSdAD/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyDownRelation/</code>	<code>\GTSdADR/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyDown/</code>	<code>\GTSnDAD/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyDownRelation/</code>	<code>\GTSnDADR/</code>	$\circ$	$u \circ v$
<code>\GTSincidence/</code>	<code>\GTSi/</code>	$\circ$	$v \circ e$
<code>\GTSincidenceRelation/</code>	<code>\GTSiR/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedIncidence/</code>	<code>\GTSnI/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedIncidenceRelation/</code>	<code>\GTSnIR/</code>	$\circ$	$v \circ e$
<code>\GTSoutIncidence/</code>	<code>\GTSoI/</code>	$\circ$	$v \circ e$
<code>\GTSoutIncidenceRelation/</code>	<code>\GTSoIR/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedOutIncidence/</code>	<code>\GTSnOI/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedOutIncidenceRelation/</code>	<code>\GTSnOIR/</code>	$\circ$	$v \circ e$
<code>\GTSinIncidence/</code>	<code>\GTSiI/</code>	$\circ$	$v \circ e$
<code>\GTSinIncidenceRelation/</code>	<code>\GTSiIR/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedInIncidence/</code>	<code>\GTSnII/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedInIncidenceRelation/</code>	<code>\GTSnIIR/</code>	$\circ$	$v \circ e$
Long name	Short name	Result	In context
<code>\GTSadjacencyMonochrome/</code>	<code>\GTSaM/</code>	$\circ$	$u \circ v$
<code>\GTSadjacencyMonochromeRelation/</code>	<code>\GTSaMR/</code>	$\circ$	$u \circ v$
<code>\GTSunadjacencyMonochrome/</code>	<code>\GTSuM/</code>	$\circ$	$u \circ v$
<code>\GTSunadjacencyMonochromeRelation/</code>	<code>\GTSuMR/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyUpMonochrome/</code>	<code>\GTSdAUM/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyUpMonochromeRelation/</code>	<code>\GTSdAUMR/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyUpMonochrome/</code>	<code>\GTSnDAUM/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyUpMonochromeRelation/</code>	<code>\GTSnDAUMR/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyDownMonochrome/</code>	<code>\GTSdADM/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyDownMonochromeRelation/</code>	<code>\GTSdADMR/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyDownMonochrome/</code>	<code>\GTSnDADM/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyDownMonochromeRelation/</code>	<code>\GTSnDADMR/</code>	$\circ$	$u \circ v$
<code>\GTSincidenceMonochrome/</code>	<code>\GTSiM/</code>	$\circ$	$v \circ e$
<code>\GTSincidenceMonochromeRelation/</code>	<code>\GTSiMR/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedIncidenceMonochrome/</code>	<code>\GTSnIM/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedIncidenceMonochromeRelation/</code>	<code>\GTSnIMR/</code>	$\circ$	$v \circ e$
<code>\GTSoutIncidenceMonochrome/</code>	<code>\GTSoIM/</code>	$\circ$	$v \circ e$
<code>\GTSoutIncidenceMonochromeRelation/</code>	<code>\GTSoIMR/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedOutIncidenceMonochrome/</code>	<code>\GTSnOIM/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedOutIncidenceMonochromeRelation/</code>	<code>\GTSnOIMR/</code>	$\circ$	$v \circ e$
<code>\GTSinIncidenceMonochrome/</code>	<code>\GTSiIM/</code>	$\circ$	$v \circ e$
<code>\GTSinIncidenceMonochromeRelation/</code>	<code>\GTSiIMR/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedInIncidenceMonochrome/</code>	<code>\GTSnIIM/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedInIncidenceMonochromeRelation/</code>	<code>\GTSnIIMR/</code>	$\circ$	$v \circ e$

\GTSsetMainColor{blue}			
Long name	Short name	Result	In context
\GTSadjacency/	\GTSa/		u v
\GTSadjacencyRelation/	\GTSaR/		u v
\GTSunadjacency/	\GTSu/		u v
\GTSunadjacencyRelation/	\GTSuR/		u v
\GTSdirectedAdjacencyUp/	\GTSdAU/		u v
\GTSdirectedAdjacencyUpRelation/	\GTSdAUR/		u v
\GTSnegatedDirectedAdjacencyUp/	\GTSnDAU/		u v
\GTSnegatedDirectedAdjacencyUpRelation/	\GTSnDAUR/		u v
\GTSdirectedAdjacencyDown/	\GTSdAD/		u v
\GTSdirectedAdjacencyDownRelation/	\GTSdADR/		u v
\GTSnegatedDirectedAdjacencyDown/	\GTSnDAD/		u v
\GTSnegatedDirectedAdjacencyDownRelation/	\GTSnDADR/		u v
\GTSincidence/	\GTSi/		v e
\GTSincidenceRelation/	\GTSiR/		v e
\GTSnegatedIncidence/	\GTSnI/		v e
\GTSnegatedIncidenceRelation/	\GTSnIR/		v e
\GTSoutIncidence/	\GTSOI/		v e
\GTSoutIncidenceRelation/	\GTSOIR/		v e
\GTSnegatedOutIncidence/	\GTSnOI/		v e
\GTSnegatedOutIncidenceRelation/	\GTSnOIR/		v e
\GTSinIncidence/	\GTSiI/		v e
\GTSinIncidenceRelation/	\GTSiIR/		v e
\GTSnegatedInIncidence/	\GTSnII/		v e
\GTSnegatedInIncidenceRelation/	\GTSnIIR/		v e

2.2 Detailed usage

First you need to add in your preamble: `\usepackage{graph-theory-symbol}`.

All symbols macros were designed with ease of use in mind: they are called with a trailing slash, they can be called in text mode or in math mode, a variant with relation spacing for math mode exists with suffix “relation”.

`\GTS@newcommand` This macro will enforce that the other macros are called with a trailing slash, avoiding problems with spaces after macro call.

Arguments:

`{\langle commandname \rangle}` The name of the command to create.

`{\langle commandcode \rangle}` The code of the command to create.

`\GTSsetMainColor` This macro will set the main color used in GraphTheorySymbol. This color has initial value “black”.

Argument:

`{\langle color \rangle}` The color that must be used by GraphTheorySymbol afterward.

`\GTSsetSecondColor` This macro will set the second color used in GraphTheorySymbol. This color has initial value “gray”.

Argument:

`{\langle color \rangle}` The color that must be used by GraphTheorySymbol afterward.

`\GTS@a` This macro will draw any of the adjacency symbols if the correct arguments are given. It is not expected to be used directly. See the macros derived from it and `\GTS@b`: `\GTSxadjacency`, `\GTSxadj`, and `\GTSxa`. Arguments:

`{\langle mainColor \rangle}` If non-empty, this color will be used as main color.

`{\langle secondColor \rangle}` If non-empty, this color will be used as second color.

`{\langle drawSlash \rangle}` Non-empty implies that a negation will be drawn.

`{\langle drawUpArrow \rangle}` Non-empty implies that an up-arrow will be drawn.

`{\langle drawDownArrow \rangle}` Non-empty implies that a down-arrow will be drawn.

`\GTS@b` This macro builds on top of `\GTS@a` to handle mathematical relation spacing if sixth argument is not empty. It is not expected to be used directly. See the macros derived from it and `\GTS@a`: `\GTSxadjacency`, `\GTSxadj`, and `\GTSxa`. Arguments:

`{\langle mainColor \rangle}` If non-empty, this color will be used as main color.

`{\langle secondColor \rangle}` If non-empty, this color will be used as second color.

`{\langle drawSlash \rangle}` Non-empty implies that a negation will be drawn.

`{\langle drawUpArrow \rangle}` Non-empty implies that an up-arrow will be drawn.

`{\langle drawDownArrow \rangle}` Non-empty implies that a down-arrow will be drawn.

`{\langle relation \rangle}` Non-empty implies that relation spacing will be added.

`\GTSxa` Versatile macros for adjacency that can be given options. Argument:

`\GTSxadj` `[\langle optionsList \rangle]` Sets the following options if not empty:

`\GTSxadjacency` `mainColor=color1` or `mainC=color1` or `mColor=color1` or `mC=color1`,
`secondColor=color2` or `secondC=color2` or `sColor=color2` or `sC=color2`,
`monochrome` or `mon` or `m` (equivalent to `mC=black,sC=black`),
`negated` or `not` or `n`,
`up` or `u`,
`down` or `d`,
`relation` or `rel` or `r`.

`\GTSadjacency` This macro will draw an adjacency symbol. A first drawing didn’t include the small dots in the middle of the two circles corresponding to the graph vertices. But it was too similar to “spoon” and `multimap` symbols, and could be confused with a graph with only two vertices and one edge. Hence, these dots were added and they have the nice property to remind of the classical textual notation of adjacency relation as `Adj(.,.)`.

`A.\GTSadjacency/A.` or `A.\GTSadj/A.` or `A.\GTSa/A.` will yield $A \mathrel{\mathbin{\mathcal{A}}} A$.

`u \GTSadjacency/ v` or `u \GTSadj/ v` or `u \GTSa/ v` will yield $u \mathrel{\mathbin{\mathcal{A}}} v$.

`$u \GTSadjacency/ v$` or `$u \GTSadj/ v$` or `$u \GTSa/ v$` will yield $u \mathrel{\mathbin{\mathcal{A}}} v$.

`\GTSadjacencyRelation` This macro will draw an adjacency symbol.

`\GTSadjRel` `$u \GTSadjacencyRelation/v$` or `$u \GTSadjRel/v$` or `$u \GTSaR/v$` will yield $u \mathrel{\mathbin{\mathcal{A}}} v$.

`\GTSaR` This macro will draw an unadjacency symbol.

`\GTSunadjacency` `A.\GTSunadjacency/A.` or `A.\GTSunadj/A.` or `A.\GTSu/A.` will yield $A \mathrel{\mathbin{\mathcal{A}}} A$.

`\GTSunadj` `u \GTSunadjacency/ v` or `u \GTSunadj/ v` or `u \GTSu/ v` will yield $u \mathrel{\mathbin{\mathcal{A}}} v$.

`\GTSu` `$u \GTSunadjacency/ v$` or `$u \GTSunadj/ v$` or `$u \GTSu/ v$` will yield $u \mathrel{\mathbin{\mathcal{A}}} v$.

`\GTSunadjacencyRelation` This macro will draw an unadjacency symbol.

`\GTSunadjRel` `$u \GTSunadjacencyRelation/v$` or `$u \GTSunadjRel/v$` or `$u \GTSuR/v$` will yield $u \mathrel{\mathbin{\mathcal{A}}} v$.

`\GTSuR`

`\GTSdirectedAdjacencyUp` This macro will draw a directed adjacency upward symbol. The operand on the left of the symbol should be considered the source of the arc; the operand on the right of the symbol should be considered the target of the arc.
`\GTSdirAdjUp` $A.\overset{\circ}{\text{A}}$. or $A.\overset{\circ}{\text{A}}$. will yield $A.\overset{\circ}{\text{A}}$.
`\GTSdAU` $u \overset{\circ}{\text{v}}$ or $u \overset{\circ}{\text{v}}$ or $u \overset{\circ}{\text{v}}$ will yield $u \overset{\circ}{\text{v}}$.
 $\$u \overset{\circ}{\text{v}}$ or $\$u \overset{\circ}{\text{v}}$ or $\$u \overset{\circ}{\text{v}}$ will yield $u \overset{\circ}{\text{v}}$.

`\GTSdirectedAdjacencyUp-` This macro will draw a directed adjacency upward symbol with relation spacing in math mode.
`Relation` $\$u \overset{\circ}{\text{v}}$ or $\$u \overset{\circ}{\text{v}}$ or $\$u \overset{\circ}{\text{v}}$
`\GTSdirAdjUpRel` will yield $u \overset{\circ}{\text{v}}$.
`\GTSdAUR`

`\GTSnegatedDirected-` This macro will draw a negated directed adjacency upward symbol.
`AdjacencyUp` $A.\overset{\circ}{\text{A}}$. or $A.\overset{\circ}{\text{A}}$. or $A.\overset{\circ}{\text{A}}$.
`\GTSnegDirAdjUp` will yield $A.\overset{\circ}{\text{A}}$.
`\GTSnDAU` $u \overset{\circ}{\text{v}}$ or $u \overset{\circ}{\text{v}}$ or $u \overset{\circ}{\text{v}}$ will yield $u \overset{\circ}{\text{v}}$.
 $\$u \overset{\circ}{\text{v}}$ or $\$u \overset{\circ}{\text{v}}$ or $\$u \overset{\circ}{\text{v}}$ will yield $u \overset{\circ}{\text{v}}$.

`\GTSnegatedDirected-` This macro will draw a negated directed adjacency upward symbol with relation spacing in math mode.
`AdjacencyUpRelation` $\$u \overset{\circ}{\text{v}}$ or $\$u \overset{\circ}{\text{v}}$ or $\$u \overset{\circ}{\text{v}}$
`\GTSnegDirAdjUpRel` will yield $u \overset{\circ}{\text{v}}$.
`\GTSnDAUR`

`\GTSdirectedAdjacencyDown` This macro will draw a directed adjacency downward symbol. The operand on the left of the symbol should be considered the target of the arc; the operand on the right of the symbol should be considered the source of the arc.
`\GTSdirAdjDown` $A.\underset{\circ}{\text{A}}$. or $A.\underset{\circ}{\text{A}}$. or $A.\underset{\circ}{\text{A}}$. will yield $A.\underset{\circ}{\text{A}}$.
 $u \underset{\circ}{\text{v}}$ or $u \underset{\circ}{\text{v}}$ or $u \underset{\circ}{\text{v}}$ will yield $u \underset{\circ}{\text{v}}$.
 $\$u \underset{\circ}{\text{v}}$ or $\$u \underset{\circ}{\text{v}}$ or $\$u \underset{\circ}{\text{v}}$ will yield $u \underset{\circ}{\text{v}}$.

`\GTSdirectedAdjacencyDown-` This macro will draw a directed adjacency downward symbol with relation spacing in math mode.
`Relation` $\$u \underset{\circ}{\text{v}}$ or $\$u \underset{\circ}{\text{v}}$ or $\$u \underset{\circ}{\text{v}}$
`\GTSdirAdjDownRel` will yield $u \underset{\circ}{\text{v}}$.
`\GTSdADR`

`\GTSnegatedDirected-` This macro will draw a negated directed adjacency downward symbol.
`AdjacencyDown` $A.\underset{\circ}{\text{A}}$. or $A.\underset{\circ}{\text{A}}$. or $A.\underset{\circ}{\text{A}}$.
`\GTSnegDirAdjDown` will yield $A.\underset{\circ}{\text{A}}$.
`\GTSnDAD` $u \underset{\circ}{\text{v}}$ or $u \underset{\circ}{\text{v}}$ or $u \underset{\circ}{\text{v}}$ will yield $u \underset{\circ}{\text{v}}$.
 $\$u \underset{\circ}{\text{v}}$ or $\$u \underset{\circ}{\text{v}}$ or $\$u \underset{\circ}{\text{v}}$ will yield $u \underset{\circ}{\text{v}}$.

`\GTSnegatedDirected-` This macro will draw a negated directed adjacency downward symbol with relation spacing in math mode.
`AdjacencyDownRelation` $\$u \underset{\circ}{\text{v}}$ or $\$u \underset{\circ}{\text{v}}$ or $\$u \underset{\circ}{\text{v}}$
`\GTSnegDirAdjDownRel` will yield $u \underset{\circ}{\text{v}}$.
`\GTSnDADR`

`\GTSincidence` This macro will draw an incidence symbol. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.
`\GTSi` $A.\downarrow$. or $A.\downarrow$. will yield $A.\downarrow$.
 $v \downarrow e$ or $v \downarrow e$ will yield $v \downarrow e$.
 $\$v \downarrow e$ or $\$v \downarrow e$ will yield $v \downarrow e$.

`\GTSincidenceRelation` This macro will draw an incidence symbol with relation spacing in math mode.
`\GTSiR` $v \downarrow e$ or $v \downarrow e$ will yield $v \downarrow e$.
`\GTSnDIR`

`\GTSnegatedIncidence` This macro will draw a negated incidence symbol.
`\GTSnI` $A.\nmid$. or $A.\nmid$. will yield $A.\nmid$.
 $v \nmid e$ or $v \nmid e$ will yield $v \nmid e$.
 $\$v \nmid e$ or $\$v \nmid e$ will yield $v \nmid e$.

`\GTSnegatedIncidence-` This macro will draw a negated incidence symbol with relation spacing in math mode.
`Relation` $v \nmid e$ or $v \nmid e$ will yield $v \nmid e$.
`\GTSnIR`

`\GTSoutIncidence` This macro will draw a directed incidence upward (out(ward) incidence) symbol. The
`\GTSOI`

operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

$A.\backslash\text{GTSoutIncidence}/A.$ or $A.\backslash\text{GTSOI}/A.$ will yield $A.\overset{\circ}{\circ}A.$

$v\ \backslash\text{GTSoutIncidence}/\ e$ or $v\ \backslash\text{GTSOI}/\ e$ will yield $v\ \overset{\circ}{\circ}e.$

$\$v\ \backslash\text{GTSoutIncidence}/\ e\$$ or $\$v\ \backslash\text{GTSOI}/\ e\$$ will yield $v\overset{\circ}{\circ}e.$

$\backslash\text{GTSoutIncidenceRelation}$ This macro will draw a directed incidence upward (out(ward) incidence) symbol with
 $\backslash\text{GTSOIR}$ relation spacing in math mode.

$\$v\backslash\text{GTSoutIncidenceRelation}/e\$$ or $\$v\backslash\text{GTSOIR}/e\$$ will yield $v\ \overset{\circ}{\circ}e.$

$\backslash\text{GTSnegatedOutIncidence}$ This macro will draw a negated directed incidence upward (out(ward) incidence) symbol.

$\backslash\text{GTSnOI}$ $A.\backslash\text{GTSnegatedOutIncidence}/A.$ or $A.\backslash\text{GTSnOI}/A.$ will yield $A.\overset{\circ}{\circ}A.$

$v\ \backslash\text{GTSnegatedOutIncidence}/\ e$ or $v\ \backslash\text{GTSnOI}/\ e$ will yield $v\ \overset{\circ}{\circ}e.$

$\$v\ \backslash\text{GTSnegatedOutIncidence}/\ e\$$ or $\$v\ \backslash\text{GTSnOI}/\ e\$$ will yield $v\overset{\circ}{\circ}e.$

$\backslash\text{GTSnegatedOutIncidence-}$ This macro will draw a negated directed incidence upward (out(ward) incidence)
 Relation symbol with relation spacing in math mode.

$\backslash\text{GTSnOIR}$ $\$v\backslash\text{GTSnegatedOutIncidenceRelation}/e\$$ or $\$v\backslash\text{GTSnOIR}/e\$$ will yield $v\ \overset{\circ}{\circ}e.$

$\backslash\text{GTSinIncidence}$ This macro will draw a directed incidence downward (in(ward) incidence) symbol. The
 $\backslash\text{GTSiI}$ operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

$A.\backslash\text{GTSinIncidence}/A.$ or $A.\backslash\text{GTSiI}/A.$ will yield $A.\overset{\circ}{\circ}A.$

$v\ \backslash\text{GTSinIncidence}/\ e$ or $v\ \backslash\text{GTSiI}/\ e$ will yield $v\ \overset{\circ}{\circ}e.$

$\$v\ \backslash\text{GTSinIncidence}/\ e\$$ or $\$v\ \backslash\text{GTSiI}/\ e\$$ will yield $v\overset{\circ}{\circ}e.$

$\backslash\text{GTSinIncidenceRelation}$ This macro will draw a directed incidence downward (in(ward) incidence) symbol with
 $\backslash\text{GTSiIR}$ relation spacing in math mode.

$\$v\backslash\text{GTSinIncidenceRelation}/e\$$ or $\$v\backslash\text{GTSiIR}/e\$$ will yield $v\ \overset{\circ}{\circ}e.$

$\backslash\text{GTSnegatedInIncidence}$ This macro will draw a negated directed incidence downward (in(ward) incidence) sym-
 $\backslash\text{GTSnII}$ bol.

$A.\backslash\text{GTSnegatedInIncidence}/A.$ or $A.\backslash\text{GTSnII}/A.$ will yield $A.\overset{\circ}{\circ}A.$

$v\ \backslash\text{GTSnegatedInIncidence}/\ e$ or $v\ \backslash\text{GTSnII}/\ e$ will yield $v\ \overset{\circ}{\circ}e.$

$\$v\ \backslash\text{GTSnegatedInIncidence}/\ e\$$ or $\$v\ \backslash\text{GTSnII}/\ e\$$ will yield $v\overset{\circ}{\circ}e.$

$\backslash\text{GTSnegatedInIncidence-}$ This macro will draw a negated directed incidence downward (in(ward) incidence)
 Relation symbol with relation spacing in math mode.

$\backslash\text{GTSnIIR}$ $\$v\backslash\text{GTSnegatedInIncidenceRelation}/e\$$ or $\$v\backslash\text{GTSnIIR}/e\$$ will yield $v\ \overset{\circ}{\circ}e.$

$\backslash\text{GTSadjacencyMonochrome}$ This macro will draw a monochrome adjacency symbol.

$\backslash\text{GTSaM}$ $A.\backslash\text{GTSadjacencyMonochrome}/A.$ or $A.\backslash\text{GTSaM}/A.$ will yield $A.\overset{\circ}{\circ}A.$

$u\ \backslash\text{GTSadjacencyMonochrome}/\ v$ or $u\ \backslash\text{GTSaM}/\ v$ will yield $u\ \overset{\circ}{\circ}v.$

$\$u\ \backslash\text{GTSadjacencyMonochrome}/\ v\$$ or $\$u\ \backslash\text{GTSaM}/\ v\$$ will yield $u\overset{\circ}{\circ}v.$

$\backslash\text{GTSadjacencyMonochrome-}$ This macro will draw a monochrome adjacency symbol with relation spacing in math
 Relation mode.

$\backslash\text{GTSaMR}$ $\$u\backslash\text{GTSadjacencyMonochromeRelation}/v\$$ or $\$u\backslash\text{GTSaMR}/v\$$ will yield $u\ \overset{\circ}{\circ}v.$

$\backslash\text{GTSunadjacencyMonochrome}$ This macro will draw a monochrome unadjacency symbol.

$\backslash\text{GTSuM}$ $A.\backslash\text{GTSunadjacencyMonochrome}/A.$ or $A.\backslash\text{GTSuM}/A.$ will yield $A.\overset{\circ}{\circ}A.$

$u\ \backslash\text{GTSunadjacencyMonochrome}/\ v$ or $u\ \backslash\text{GTSuM}/\ v$ will yield $u\ \overset{\circ}{\circ}v.$

$\$u\ \backslash\text{GTSunadjacencyMonochrome}/\ v\$$ or $\$u\ \backslash\text{GTSuM}/\ v\$$ will yield $u\overset{\circ}{\circ}v.$

$\backslash\text{GTSunadjacencyMonochrome-}$ This macro will draw a monochrome unadjacency symbol with relation spacing in
 Relation math mode.

$\backslash\text{GTSuMR}$ $\$u\backslash\text{GTSunadjacencyMonochromeRelation}/v\$$ or $\$u\backslash\text{GTSuMR}/v\$$ will yield $u\ \overset{\circ}{\circ}v.$

$\backslash\text{GTSdirectedAdjacencyUp-}$ This macro will draw a monochrome directed adjacency upward symbol. The operand
 Monochrome on the left of the symbol should be considered the source of the arc; the operand on the right of the symbol should be considered the target of the arc.

$A.\backslash\text{GTSdirectedAdjacencyUpMonochrome}/A.$ or $A.\backslash\text{GTSdAUM}/A.$ will yield $A.\overset{\circ}{\circ}A.$

$u\ \backslash\text{GTSdirectedAdjacencyUpMonochrome}/\ v$ or $u\ \backslash\text{GTSdAUM}/\ v$ will yield $u\ \overset{\circ}{\circ}v.$

$\$u\ \backslash\text{GTSdirectedAdjacencyUpMonochrome}/\ v\$$ or $\$u\ \backslash\text{GTSdAUM}/\ v\$$ will yield $u\overset{\circ}{\circ}v.$

$\backslash\text{GTSdirectedAdjacencyUp-}$ This macro will draw a monochrome directed adjacency upward symbol with relation
 $\text{MonochromeRelation}$ spacing in math mode.

$\backslash\text{GTSdAUMR}$ $\$u\backslash\text{GTSdirectedAdjacencyUpMonochromeRelation}/v\$$ or $\$u\backslash\text{GTSdAUMR}/v\$$ will yield $u\ \overset{\circ}{\circ}v.$

$\backslash\text{GTSnegatedDirected-}$ This macro will draw a monochrome negated directed adjacency upward symbol.

$\text{AdjacencyUpMonochrome}$ $A.\backslash\text{GTSnegatedDirectedAdjacencyUpMonochrome}/A.$ or $A.\backslash\text{GTSnDAUM}/A.$ will yield $A.\overset{\circ}{\circ}A.$

$\backslash\text{GTSnDAUM}$ $u\ \backslash\text{GTSnegatedDirectedAdjacencyUpMonochrome}/\ v$ or $u\ \backslash\text{GTSnDAUM}/\ v$ will yield $u\ \overset{\circ}{\circ}v.$

$\$u\ \backslash\text{GTSnegatedDirectedAdjacencyUpMonochrome}/\ v\$$ or $\$u\ \backslash\text{GTSnDAUM}/\ v\$$ will yield $u\overset{\circ}{\circ}v.$

`\GTSnegatedDirectedAdjacencyUpMonochromeRelation` This macro will draw a monochrome negated directed adjacency upward symbol with relation spacing in math mode. `\GTSnDAUMR/v$` or `$u\GTSnDAUMR/v$` will yield $u \overset{\circ}{\frown} v$.

`\GTSdirectedAdjacencyDownMonochrome` This macro will draw a monochrome directed adjacency downward symbol. The operand on the left of the symbol should be considered the target of the arc; the operand on the right of the symbol should be considered the source of the arc. `A.\GTSdirectedAdjacencyDownMonochrome/A.` or `A.\GTSdADM/A.` will yield $A \overset{\circ}{\smile} A$. `u \GTSdirectedAdjacencyDownMonochrome/ v` or `u \GTSdADM/ v` will yield $u \overset{\circ}{\smile} v$. `$u \GTSdirectedAdjacencyDownMonochrome/ v$` or `$u \GTSdADM/ v$` will yield $u \overset{\circ}{\smile} v$.

`\GTSdirectedAdjacencyDownMonochromeRelation` This macro will draw a monochrome directed adjacency downward symbol with relation spacing in math mode. `\GTSdADMR` `$u\GTSdirectedAdjacencyDownMonochromeRelation/v$` or `$u\GTSdADMR/v$` will yield $u \overset{\circ}{\smile} v$.

`\GTSnegatedDirectedAdjacencyDownMonochrome` This macro will draw a monochrome negated directed adjacency downward symbol. `A.\GTSnegatedDirectedAdjacencyDownMonochrome/A.` or `A.\GTSnDADM/A.` will yield $A \overset{\circ}{\frown} A$. `u \GTSnegatedDirectedAdjacencyDownMonochrome/ v` or `u \GTSnDADM/ v` will yield $u \overset{\circ}{\frown} v$. `$u \GTSnegatedDirectedAdjacencyDownMonochrome/ v$` or `$u \GTSnDADM/ v$` will yield $u \overset{\circ}{\frown} v$.

`\GTSnegatedDirectedAdjacencyDownMonochromeRelation` This macro will draw a monochrome negated directed adjacency downward symbol with relation spacing in math mode. `\GTSnDADMR` `$u\GTSnegatedDirectedAdjacencyDownMonochromeRelation/v$` or `$u\GTSnDADMR/v$` will yield $u \overset{\circ}{\frown} v$.

`\GTSincidenceMonochrome` This macro will draw a monochrome incidence symbol. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident. `A.\GTSincidenceMonochrome/A.` or `A.\GTSiM/A.` will yield $A \downarrow A$. `v \GTSincidenceMonochrome/ e` or `v \GTSiM/ e` will yield $v \downarrow e$. `$v \GTSincidenceMonochrome/ e$` or `$v \GTSiM/ e$` will yield $v \downarrow e$.

`\GTSincidenceMonochromeRelation` This macro will draw a monochrome incidence symbol with relation spacing in math mode. `\GTSiMR` `$v\GTSincidenceMonochromeRelation/e$` or `$v\GTSiMR/e$` will yield $v \downarrow e$.

`\GTSnegatedIncidenceMonochrome` This macro will draw a monochrome negated incidence symbol. `A.\GTSnegatedIncidenceMonochrome/A.` or `A.\GTSnIM/A.` will yield $A \uparrow A$. `v \GTSnegatedIncidenceMonochrome/ e` or `v \GTSnIM/ e` will yield $v \uparrow e$. `$v \GTSnegatedIncidenceMonochrome/ e$` or `$v \GTSnIM/ e$` will yield $v \uparrow e$.

`\GTSnegatedIncidenceMonochromeRelation` This macro will draw a monochrome negated incidence symbol with relation spacing in math mode. `\GTSnIMR` `$v\GTSnegatedIncidenceMonochromeRelation/e$` or `$v\GTSnIMR/e$` will yield $v \uparrow e$.

`\GTSoutIncidenceMonochrome` This macro will draw a monochrome directed incidence upward (out(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident. `A.\GTSoutIncidenceMonochrome/A.` or `A.\GTSoIM/A.` will yield $A \uparrow A$. `v \GTSoutIncidenceMonochrome/ e` or `v \GTSoIM/ e` will yield $v \uparrow e$. `$v \GTSoutIncidenceMonochrome/ e$` or `$v \GTSoIM/ e$` will yield $v \uparrow e$.

`\GTSoutIncidenceMonochromeRelation` This macro will draw a monochrome directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode. `\GTSoIMR` `$v\GTSoutIncidenceMonochromeRelation/e$` or `$v\GTSoIMR/e$` will yield $v \uparrow e$.

`\GTSnegatedOutIncidenceMonochrome` This macro will draw a monochrome negated directed incidence upward (out(ward) incidence) symbol. `A.\GTSnegatedOutIncidenceMonochrome/A.` or `A.\GTSnOIM/A.` will yield $A \downarrow A$. `v \GTSnegatedOutIncidenceMonochrome/ e` or `v \GTSnOIM/ e` will yield $v \downarrow e$. `$v \GTSnegatedOutIncidenceMonochrome/ e$` or `$v \GTSnOIM/ e$` will yield $v \downarrow e$.

`\GTSnegatedOutIncidenceMonochromeRelation` This macro will draw a monochrome negated directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode. `\GTSnOIMR` `$v\GTSnegatedOutIncidenceMonochromeRelation/e$` or `$v\GTSnOIMR/e$` will yield $v \downarrow e$.

`\GTSinIncidenceMonochrome` This macro will draw a monochrome directed incidence downward (in(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident. `A.\GTSinIncidenceMonochrome/A.` or `A.\GTSiM/A.` will yield $A \downarrow A$.

`v \GTSinIncidenceMonochrome/ e` or `v \GTSiIM/ e` will yield $v \not\propto e$.
`\GTSinIncidenceMonochrome/ e$` or `\GTSiIM/ e$` will yield $v \not\propto e$.
`\GTSinIncidenceMonochrome-Relation` This macro will draw a monochrome directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.
`\GTSiIMR` `\GTSinIncidenceMonochromeRelation/e$` or `\GTSiIMR/e$` will yield $v \not\propto e$.
`\GTSnegatedInIncidenceMonochrome` This macro will draw a monochrome negated directed incidence downward (in(ward) incidence) symbol.
`\GTSniIM` `A.\GTSnegatedInIncidenceMonochrome/A.` or `A.\GTSniIM/A.` will yield $A \not\propto A$.
`v \GTSnegatedInIncidenceMonochrome/ e` or `v \GTSniIM/ e` will yield $v \not\propto e$.
`\GTSnegatedInIncidenceMonochrome/ e$` or `\GTSniIM/ e$` will yield $v \not\propto e$.
`\GTSnegatedInIncidenceMonochromeRelation` This macro will draw a monochrome negated directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.
`\GTSniIMR` `\GTSnegatedInIncidenceMonochromeRelation/e$` or `\GTSniIMR/e$` will yield $v \not\propto e$.

3 Implementation

```

1 \<package>
2 \RequirePackage{amsmath}
3 \RequirePackage{amssymb}
4 \RequirePackage{graphicx}
5 \RequirePackage{keyval}
6 \RequirePackage{fdsymbol}
7 \RequirePackage{lua-unicode-math}
8 \RequirePackage{xcolor}
9
10 \def \GTS@mainColor {black}
11 \def \GTS@secondColor {gray}
12 \def \GTS@mainColorBackup {black}
13 \def \GTS@secondColorBackup {gray}
14 \def \GTS@drawSlash {}
15 \def \GTS@drawUpArrow {}
16 \def \GTS@drawDownArrow {}
17
\GTS@newcommand This macro will enforce that the other macros are called with a trailing slash, avoiding
problems with spaces after macro call.
Arguments:
{\<commandname>} The name of the command to create.
{\<commandcode>} The code of the command to create.
18 \@ifdefinable{\GTS@newcommand}{\def\GTS@newcommand#1#2{%
19   \@ifdefinable{#1}{\def#1/{#2}}
20 }}

\GTSsetMainColor This macro will set the main color used in GraphTheorySymbol. This color has initial value
“black”.
Argument:
{\<color>} The color that must be used by GraphTheorySymbol afterward.
21 \newcommand{\GTSsetMainColor}[1]{%
22 \edef\GTS@mainColorTemp{#1}%
23 \edef\GTS@mainColorBackup{\GTS@mainColor}%
24 \edef\GTS@mainColor{\GTS@mainColorTemp}%
25 }

\GTSsetSecondColor This macro will set the second color used in GraphTheorySymbol. This color has initial
value “gray”.
Argument:
{\<color>} The color that must be used by GraphTheorySymbol afterward.
26 \newcommand{\GTSsetSecondColor}[1]{%
27 \edef\GTS@secondColorTemp{#1}%
28 \edef\GTS@secondColorBackup{\GTS@secondColor}%
29 \edef\GTS@secondColor{\GTS@secondColorTemp}%
30 }
```


`\GTS@a` This macro will draw any adjacency symbol given the correct parameters. Its use is purely technical. See the macros derived from it and `\GTS@b`: `\GTSxadjacency`, `\GTSxadj`, and `\GTSxa`.

```

31 \newcommand{\GTS@a}[5]{%
32 \if#1\empty\else%
33 \GTSsetMainColor{#1}%
34 \fi%
35 \if#2\empty\else%
36 \GTSsetSecondColor{#2}%
37 \fi%
38 \text{\ensuremath{%
39 \raisebox{-0.2ex}{\scalebox{1.5}[1.5]{%
40 %
41 \raisebox{0.7ex}{\scalebox{0.7}[0.7]{%
42 $\mathcolor{\GTS@secondColor}{\circ}$%
43 }}%
44 \hspace{-0.39ex}%
45 \raisebox{1.05ex}{\scalebox{0.1}[0.1]{%
46 $\mathcolor{\GTS@secondColor}{\bullet}$%
47 }}%
48 \hspace{-0.12ex}%
49 \raisebox{0.11ex}{\scalebox{0.1}[0.73]{%
50 $\mathcolor{\GTS@mainColor}{\smallblacksquare}$%
51 }}%
52 \hspace{-0.40ex}%
53 \raisebox{-0.3ex}{\scalebox{0.7}[0.7]{%
54 $\mathcolor{\GTS@secondColor}{\circ}$%
55 }}%
56 \hspace{-0.39ex}%
57 \raisebox{0.05ex}{\scalebox{0.1}[0.1]{%
58 $\mathcolor{\GTS@secondColor}{\bullet}$%
59 }}%
60 \hspace{0.27ex}%
if GTS@drawUpArrow empty else
61 \if#4\empty\else%
62 \hspace{-0.65ex}%
63 \raisebox{-0.18ex}{\scalebox{0.5}[0.5]{%
64 \textcolor{\GTS@mainColor}{\textasciicircum}%
65 }}%
66 \fi%
if GTS@drawDownArrow empty else
67 \if#5\empty\else%
68 \hspace{-0.65ex}%
69 \raisebox{1.37ex}{\scalebox{0.5}[-0.5]{%
70 \textcolor{\GTS@mainColor}{\textasciicircum}%
71 }}%
72 \fi%
if GTS@drawSlash empty else
73 \if#3\empty\else%
74 \hspace{-0.65ex}%
75 \raisebox{0.25ex}{\scalebox{0.6}[0.6]{%
76 \textcolor{black}{/}%
77 }}%
78 \fi%
79 %
80 }}%
81 }}%
82 \if#1\empty\else%
83 \GTSsetMainColor{\GTS@mainColorBackup}%
84 \fi%
85 \if#2\empty\else%
86 \GTSsetSecondColor{\GTS@secondColorBackup}%
87 \fi%
88 }
```

`\GTS@b` This macro will handle wrapping in mathematical relation spacing if needed. Its use is purely technical. See the macros derived from it and `\GTS@a`: `\GTSxadjacency`, `\GTSxadj`, and `\GTSxa`.

```

89 \newcommand{\GTS@b}[6]{%
90 \if#6\empty%
91 \GTS@a{#1}{#2}{#3}{#4}{#5}%
92 \else%
93 \mathrel{\GTS@a{#1}{#2}{#3}{#4}{#5}}%
94 \fi%
95 }

```

`\GTSxa` Versatile macros for adjacency that can be given options:

`\GTSxadj` `mainColor=color1` or `mainC=color1` or `mColor=color1` or `mC=color1`,
`\GTSxadjacency` `secondColor=color2` or `secondC=color2` or `sColor=color2` or `sC=color2`,
`monochrome` or `mon` or `m`,
`negated` or `not` or `n`,
`up` or `u`,
`down` or `d`,
`relation` or `rel` or `r`.

```

96 \define@key{GTSxa}{mainColor}{\def{\GTS@@mainColor}{#1}}
97 \define@key{GTSxa}{mainC}{\def{\GTS@@mainColor}{#1}}
98 \define@key{GTSxa}{mColor}{\def{\GTS@@mainColor}{#1}}
99 \define@key{GTSxa}{mC}{\def{\GTS@@mainColor}{#1}}
100 \define@key{GTSxa}{secondColor}{\def{\GTS@@secondColor}{#1}}
101 \define@key{GTSxa}{secondC}{\def{\GTS@@secondColor}{#1}}
102 \define@key{GTSxa}{sColor}{\def{\GTS@@secondColor}{#1}}
103 \define@key{GTSxa}{sC}{\def{\GTS@@secondColor}{#1}}
104 \define@key{GTSxa}{monochrome}[true]%
105 {\def{\GTS@@mainColor}{black}\def{\GTS@@secondColor}{black}}
106 \define@key{GTSxa}{mon}[true]%
107 {\def{\GTS@@mainColor}{black}\def{\GTS@@secondColor}{black}}
108 \define@key{GTSxa}{m}[true]%
109 {\def{\GTS@@mainColor}{black}\def{\GTS@@secondColor}{black}}
110 \define@key{GTSxa}{secondC}{\def{\GTS@@secondColor}{#1}}
111 \define@key{GTSxa}{sColor}{\def{\GTS@@secondColor}{#1}}
112 \define@key{GTSxa}{negated}[true]{\def{\GTS@@drawSlash}{1}}
113 \define@key{GTSxa}{not}[true]{\def{\GTS@@drawSlash}{1}}
114 \define@key{GTSxa}{n}[true]{\def{\GTS@@drawSlash}{1}}
115 \define@key{GTSxa}{up}[true]{\def{\GTS@@drawUpArrow}{1}}
116 \define@key{GTSxa}{u}[true]{\def{\GTS@@drawUpArrow}{1}}
117 \define@key{GTSxa}{down}[true]{\def{\GTS@@drawDownArrow}{1}}
118 \define@key{GTSxa}{d}[true]{\def{\GTS@@drawDownArrow}{1}}
119 \define@key{GTSxa}{relation}[true]{\def{\GTS@@relation}{1}}
120 \define@key{GTSxa}{rel}[true]{\def{\GTS@@relation}{1}}
121 \define@key{GTSxa}{r}[true]{\def{\GTS@@relation}{1}}
122
123 \newcommand{\GTSxa}[1] [] {%
124 \def{\GTS@@mainColor}{}%
125 \def{\GTS@@secondColor}{}%
126 \def{\GTS@@drawSlash}{}%
127 \def{\GTS@@drawUpArrow}{}%
128 \def{\GTS@@drawDownArrow}{}%
129 \def{\GTS@@relation}{}%
130 \setkeys{GTSxa}{#1}%
131 \GTS@b{\GTS@@mainColor}{\GTS@@secondColor}{\GTS@@drawSlash}%
132 {\GTS@@drawUpArrow}{\GTS@@drawDownArrow}{\GTS@@relation}%
133 }
134
135 \newcommand{\GTSxadj}[1] [] {\GTSxa[#1]}
136
137 \newcommand{\GTSxadjacency}[1] [] {\GTSxa[#1]}

```

`\GTSadjacency` This macro will draw an adjacency symbol.

```

\GTSadj 138 \GTS@newcommand{\GTSadjacency}{\GTS@a{}{}{}{}}
\GTSa 139 \GTS@newcommand{\GTSadj}{\GTS@a{}{}{}{}}

```

140 \GTS@newcommand{\GTSa}{\GTS@a{}{}{}{}}{}

\GTSadjacencyRelation This macro will draw an adjacency symbol with relation spacing in math mode.

\GTSadjRel 141 \GTS@newcommand{\GTSadjacencyRelation}{\mathrel{\GTSa/}}
\GTSaR 142 \GTS@newcommand{\GTSadjRel}{\mathrel{\GTSa/}}
143 \GTS@newcommand{\GTSaR}{\mathrel{\GTSa/}}

\GTSunadjacency This macro will draw an unadjacency symbol.

\GTSunadj 144 \GTS@newcommand{\GTSunadjacency}{\GTS@a{}{}{1}{}{}}{
\GTSu 145 \GTS@newcommand{\GTSunadj}{\GTS@a{}{}{1}{}{}}{
146 \GTS@newcommand{\GTSu}{\GTS@a{}{}{1}{}{}}{}

\GTSunadjacencyRelation This macro will draw an unadjacency symbol with relation spacing in math mode.

\GTSunadjRel 147 \GTS@newcommand{\GTSunadjacencyRelation}{\mathrel{\GTSu/}}
\GTSuR 148 \GTS@newcommand{\GTSunadjRel}{\mathrel{\GTSu/}}
149 \GTS@newcommand{\GTSuR}{\mathrel{\GTSu/}}

\GTSdirectedAdjacencyUp This macro will draw a directed adjacency upward symbol. The operand on the left of the
\GTSdirAdjUp symbol should be considered the source of the arc; the operand on the right of the symbol
\GTSdAU should be considered the target of the arc.

150 \GTS@newcommand{\GTSdirectedAdjacencyUp}{\GTS@a{}{}{1}{}{}}{
151 \GTS@newcommand{\GTSdirAdjUp}{\GTS@a{}{}{1}{}{}}{
152 \GTS@newcommand{\GTSdAU}{\GTS@a{}{}{1}{}{}}{}

\GTSdirectedAdjacencyUp- This macro will draw a directed adjacency upward symbol with relation spacing in math
Relation mode.

\GTSdirAdjUpRel 153 \GTS@newcommand{\GTSdirectedAdjacencyUpRelation}{\mathrel{\GTSdAU/}}
\GTSdAUR 154 \GTS@newcommand{\GTSdirAdjUpRel}{\mathrel{\GTSdAU/}}
155 \GTS@newcommand{\GTSdAUR}{\mathrel{\GTSdAU/}}

\GTSnegatedDirected- This macro will draw a negated directed adjacency upward symbol.

AdjacencyUp 156 \GTS@newcommand{\GTSnegatedDirectedAdjacencyUp}{\GTS@a{}{}{1}{1}{}{}}{
\GTSnegDirAdjUp 157 \GTS@newcommand{\GTSnegDirAdjUp}{\GTS@a{}{}{1}{1}{}{}}{
\GTSnDAU 158 \GTS@newcommand{\GTSnDAU}{\GTS@a{}{}{1}{1}{}{}}{}

\GTSnegatedDirected- This macro will draw a negated directed adjacency upward symbol with relation spacing in
AdjacencyUpRelation math mode.

\GTSnegDirAdjUpRel 159 \GTS@newcommand{\GTSnegatedDirectedAdjacencyUpRelation}{\mathrel{\GTSnDAU/}}
\GTSnDAUR 160 \GTS@newcommand{\GTSnegDirAdjUpRel}{\mathrel{\GTSnDAU/}}
161 \GTS@newcommand{\GTSnDAUR}{\mathrel{\GTSnDAU/}}

\GTSdirectedAdjacencyDown This macro will draw a directed adjacency downward symbol. The operand on the left of the
\GTSdirAdjDown symbol should be considered the target of the arc; the operand on the right of the symbol
\GTSdAD should be considered the source of the arc.

162 \GTS@newcommand{\GTSdirectedAdjacencyDown}{\GTS@a{}{}{}{1}{}{}}{
163 \GTS@newcommand{\GTSdirAdjDown}{\GTS@a{}{}{}{1}{}{}}{
164 \GTS@newcommand{\GTSdAD}{\GTS@a{}{}{}{1}{}{}}{}

\GTSdirectedAdjacencyDown- This macro will draw a directed adjacency downward symbol with relation spacing in math
Relation mode.

\GTSdirAdjDownRel 165 \GTS@newcommand{\GTSdirectedAdjacencyDownRelation}{\mathrel{\GTSdAD/}}
\GTSdADR 166 \GTS@newcommand{\GTSdirAdjDownRel}{\mathrel{\GTSdAD/}}
167 \GTS@newcommand{\GTSdADR}{\mathrel{\GTSdAD/}}

\GTSnegatedDirected- This macro will draw a negated directed adjacency downward symbol.

AdjacencyDown 168 \GTS@newcommand{\GTSnegatedDirectedAdjacencyDown}{\GTS@a{}{}{1}{}{1}{}{}}{
\GTSnegDirAdjDown 169 \GTS@newcommand{\GTSnegDirAdjDown}{\GTS@a{}{}{1}{}{1}{}{}}{
\GTSnDAD 170 \GTS@newcommand{\GTSnDAD}{\GTS@a{}{}{1}{}{1}{}{}}{}

\GTSnegatedDirected- This macro will draw a negated directed adjacency downward symbol with relation spacing
AdjacencyDownRelation in math mode.

\GTSnegDirAdjDownRel 171 \GTS@newcommand{\GTSnegatedDirectedAdjacencyDownRelation}{\mathrel{\GTSnDAD/}}
\GTSnDADR 172 \GTS@newcommand{\GTSnegDirAdjDownRel}{\mathrel{\GTSnDAD/}}
173 \GTS@newcommand{\GTSnDADR}{\mathrel{\GTSnDAD/}}

`\GTSincidence` This macro will draw an incidence symbol. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.

```

174 \GTS@newcommand{\GTSincidence}{%
175 \text{\ensuremath{%
176 \raisebox{0ex}{\scalebox{1.8}[1.8]{%
177 %
178 \raisebox{-0.3ex}{\scalebox{0.7}[0.7]{%
179 $\mathcolor{\GTS@secondColor}{\circ}$%
180 }}%
181 \hspace{-0.39ex}%
182 \raisebox{0.05ex}{\scalebox{0.1}[0.1]{%
183 $\mathcolor{\GTS@secondColor}{\bullet}$%
184 }}%
185 \hspace{-0.12ex}%
186 \raisebox{0.11ex}{\scalebox{0.1}[0.73]{%
187 $\mathcolor{\GTS@secondColor}{\smallblacksquare}$%
188 }}%
189 \hspace{-0.18ex}%
190 \raisebox{0.15ex}{\scalebox{0.2}[0.2]{%
191 $\mathcolor{\GTS@mainColor}{\smallblacksquare}$%
192 }}%
193 \hspace{0.2ex}%
194 \if\GTS@drawUpArrow\empty\else%
195 \hspace{-0.65ex}%
196 \raisebox{-0.18ex}{\scalebox{0.5}[0.5]{%
197 \textcolor{\GTS@secondColor}{\textasciicircum}%
198 }}%
199 \fi%
200 \if\GTS@drawDownArrow\empty\else%
201 \hspace{-0.65ex}%
202 \raisebox{1.37ex}{\scalebox{0.5}[-0.5]{%
203 \textcolor{\GTS@secondColor}{\textasciicircum}%
204 }}%
205 \fi%
206 \if\GTS@drawSlash\empty\else%
207 \hspace{-0.65ex}%
208 \raisebox{0.1ex}{\scalebox{0.5}[0.5]{%
209 \textcolor{black}{/}%
210 }}%
211 \fi%
212 %
213 }}%
214 }}%
215 }
216
217 \GTS@newcommand{\GTSi}{\GTSincidence/}

```

`\GTSincidenceRelation` This macro will draw an incidence symbol with relation spacing in math mode.

```

\GTSiR 218 \GTS@newcommand{\GTSincidenceRelation}{\mathrel{\GTSincidence/}}
219 \GTS@newcommand{\GTSiR}{\GTSincidenceRelation/}

```

`\GTSnegatedIncidence` This macro will draw a negated incidence symbol.

```

\GTSnI 220 \GTS@newcommand{\GTSnegatedIncidence}{%
221 \def\GTS@drawSlash{1}%
222 \GTSincidence/%
223 \def\GTS@drawSlash{}%
224 }
225
226 \GTS@newcommand{\GTSnI}{\GTSnegatedIncidence/}

```

`\GTSnegatedIncidenceRelation` This macro will draw a negated incidence symbol with relation spacing in math mode.

```

\GTSnIR 227 \GTS@newcommand{\GTSnegatedIncidenceRelation}{%
228 \mathrel{\GTSnegatedIncidence/}%
229 }
230 \GTS@newcommand{\GTSnIR}{\GTSnegatedIncidenceRelation/}

```

`\GTSoutIncidence` This macro will draw a directed incidence upward (out(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

```

231 \GTS@newcommand{\GTSoutIncidence}{%
232 \def\GTS@drawUpArrow{1}%
233 \GTSincidence/%
234 \def\GTS@drawUpArrow{}%
235 }
236
237 \GTS@newcommand{\GTSoI}{\GTSoutIncidence/}

```

`\GTSoutIncidenceRelation` This macro will draw a directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

```

238 \GTS@newcommand{\GTSoutIncidenceRelation}{%
239 \mathrel{\GTSoutIncidence/}%
240 }
241 \GTS@newcommand{\GTSoIR}{\GTSoutIncidenceRelation/}

```

`\GTSnegatedOutIncidence` This macro will draw a negated directed incidence upward (out(ward) incidence) symbol.

```

242 \GTS@newcommand{\GTSnegatedOutIncidence}{%
243 \def\GTS@drawSlash{1}%
244 \GTSoutIncidence/%
245 \def\GTS@drawSlash{}%
246 }
247
248 \GTS@newcommand{\GTSnOI}{\GTSnegatedOutIncidence/}

```

`\GTSnegatedOutIncidenceRelation` This macro will draw a negated directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

```

249 \GTS@newcommand{\GTSnegatedOutIncidenceRelation}{%
250 \mathrel{\GTSnegatedOutIncidence/}%
251 }
252 \GTS@newcommand{\GTSnOIR}{\GTSnegatedOutIncidenceRelation/}

```

`\GTSinIncidence` This macro will draw a directed incidence downward (in(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

```

253 \GTS@newcommand{\GTSinIncidence}{%
254 \def\GTS@drawDownArrow{1}%
255 \GTSincidence/%
256 \def\GTS@drawDownArrow{}%
257 }
258
259 \GTS@newcommand{\GTSiI}{\GTSinIncidence/}

```

`\GTSinIncidenceRelation` This macro will draw a directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

```

260 \GTS@newcommand{\GTSinIncidenceRelation}{%
261 \mathrel{\GTSinIncidence/}%
262 }
263 \GTS@newcommand{\GTSiIR}{\GTSinIncidenceRelation/}

```

`\GTSnegatedInIncidence` This macro will draw a negated directed incidence downward (in(ward) incidence) symbol.

```

264 \GTS@newcommand{\GTSnegatedInIncidence}{%
265 \def\GTS@drawSlash{1}%
266 \GTSinIncidence/%
267 \def\GTS@drawSlash{}%
268 }
269
270 \GTS@newcommand{\GTSnII}{\GTSnegatedInIncidence/}

```

`\GTSnegatedInIncidenceRelation` This macro will draw a negated directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

```

271 \GTS@newcommand{\GTSnegatedInIncidenceRelation}{%

```

```

272 \mathrel{\GTSnegatedInIncidence/}%
273 }
274 \GTS@newcommand{\GTSnIIR}{\GTSnegatedInIncidenceRelation/}

```

`\GTSadjacencyMonochrome` This macro will draw a monochrome adjacency symbol.

```

\GTSaM 275 \GTS@newcommand{\GTSadjacencyMonochrome}{%
276 \GTSsetMainColor{black}%
277 \GTSsetSecondColor{black}%
278 \GTSadjacency/%
279 \GTSsetMainColor{\GTS@mainColorBackup}%
280 \GTSsetSecondColor{\GTS@secondColorBackup}%
281 }
282
283 \GTS@newcommand{\GTSaM}{\GTSadjacencyMonochrome/}

```

`\GTSadjacencyMonochrome-` This macro will draw a monochrome adjacency symbol with relation spacing in math mode.

```

Relation 284 \GTS@newcommand{\GTSadjacencyMonochromeRelation}{%
\GTSaMR 285 \mathrel{\GTSadjacencyMonochrome/}%
286 }
287 \GTS@newcommand{\GTSaMR}{\GTSadjacencyMonochromeRelation/}

```

`\GTSunadjacencyMonochrome` This macro will draw a monochrome unadjacency symbol.

```

\GTSuM 288 \GTS@newcommand{\GTSunadjacencyMonochrome}{%
289 \def\GTS@drawSlash{1}%
290 \GTSadjacencyMonochrome/%
291 \def\GTS@drawSlash{}%
292 }
293
294 \GTS@newcommand{\GTSuM}{\GTSunadjacencyMonochrome/}

```

`\GTSunadjacencyMonochrome-` This macro will draw a monochrome unadjacency symbol with relation spacing in math mode.

```

Relation 295 \GTS@newcommand{\GTSunadjacencyMonochromeRelation}{%
\GTSuMR 296 \mathrel{\GTSunadjacencyMonochrome/}%
297 }
298 \GTS@newcommand{\GTSuMR}{\GTSunadjacencyMonochromeRelation/}

```

`\GTSdirectedAdjacencyUp-` This macro will draw a monochrome directed adjacency upward symbol. The operand on the left of the symbol should be considered the source of the arc; the operand on the right of the symbol should be considered the target of the arc.

```

Monochrome 299 \GTS@newcommand{\GTSdirectedAdjacencyUpMonochrome}{%
\GTSdAUM 300 \def\GTS@drawUpArrow{1}%
301 \GTSadjacencyMonochrome/%
302 \def\GTS@drawUpArrow{}%
303 }
304
305 \GTS@newcommand{\GTSdAUM}{\GTSdirectedAdjacencyUpMonochrome/}

```

`\GTSdirectedAdjacencyUp-` This macro will draw a monochrome directed adjacency upward symbol with relation spacing in math mode.

```

MonochromeRelation 306 \GTS@newcommand{\GTSdirectedAdjacencyUpMonochromeRelation}{%
\GTSdAUMR 307 \mathrel{\GTSdirectedAdjacencyUpMonochrome/}%
308 }
309 \GTS@newcommand{\GTSdAUMR}{\GTSdirectedAdjacencyUpMonochromeRelation/}

```

`\GTSnegatedDirected-` This macro will draw a monochrome negated directed adjacency upward symbol.

```

AdjacencyUpMonochrome 310 \GTS@newcommand{\GTSnegatedDirectedAdjacencyUpMonochrome}{%
\GTSnDAUM 311 \def\GTS@drawSlash{1}%
312 \GTSdirectedAdjacencyUpMonochrome/%
313 \def\GTS@drawSlash{}%
314 }
315
316 \GTS@newcommand{\GTSnDAUM}{\GTSnegatedDirectedAdjacencyUpMonochrome/}

```

`\GTSnegatedDirectedAdjacencyUpMonochrome-` This macro will draw a monochrome negated directed adjacency upward symbol with relation spacing in math mode.

```

Relation 317 \GTS@newcommand{\GTSnegatedDirectedAdjacencyUpMonochromeRelation}{%
\GTSnDAUMR 318 \mathrel{\GTSnegatedDirectedAdjacencyUpMonochrome/}%
319 }
320 \GTS@newcommand{\GTSnDAUMR}{%
321 \GTSnegatedDirectedAdjacencyDownMonochromeRelation/%
322 }
```

`\GTSdirectedAdjacencyDownMonochrome` This macro will draw a monochrome directed adjacency downward symbol. The operand on the left of the symbol should be considered the target of the arc; the operand on the right of the symbol should be considered the source of the arc.

```

\GTSdADM 323 \GTS@newcommand{\GTSdirectedAdjacencyDownMonochrome}{%
324 \def\GTS@drawDownArrow{1}%
325 \GTSadjacencyMonochrome/%
326 \def\GTS@drawDownArrow{}%
327 }
328
329 \GTS@newcommand{\GTSdADM}{\GTSdirectedAdjacencyDownMonochrome/}
```

`\GTSdirectedAdjacencyDownMonochromeRelation` This macro will draw a monochrome directed adjacency downward symbol with relation spacing in math mode.

```

\GTSdADMR 330 \GTS@newcommand{\GTSdirectedAdjacencyDownMonochromeRelation}{%
331 \mathrel{\GTSdirectedAdjacencyDownMonochrome/}%
332 }
333 \GTS@newcommand{\GTSdADMR}{\GTSdirectedAdjacencyDownMonochromeRelation/}
```

`\GTSnegatedDirectedAdjacencyDownMonochrome` This macro will draw a monochrome negated directed adjacency downward symbol.

```

\GTSnDADM 334 \GTS@newcommand{\GTSnegatedDirectedAdjacencyDownMonochrome}{%
335 \def\GTS@drawSlash{1}%
336 \GTSdirectedAdjacencyDownMonochrome/%
337 \def\GTS@drawSlash{}%
338 }
339
340 \GTS@newcommand{\GTSnDADM}{\GTSnegatedDirectedAdjacencyDownMonochrome/}
```

`\GTSnegatedDirectedAdjacencyDownMonochromeRelation` This macro will draw a monochrome negated directed adjacency downward symbol with relation spacing in math mode.

```

Relation 341 \GTS@newcommand{\GTSnegatedDirectedAdjacencyDownMonochromeRelation}{%
\GTSnDADMR 342 \mathrel{\GTSnegatedDirectedAdjacencyDownMonochrome/}%
343 }
344 \GTS@newcommand{\GTSnDADMR}{%
345 \GTSnegatedDirectedAdjacencyDownMonochromeRelation/%
346 }
```

`\GTSincidenceMonochrome` This macro will draw a monochrome incidence symbol. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.

```

\GTSiM 347 \GTS@newcommand{\GTSincidenceMonochrome}{%
348 \GTSsetMainColor{black}%
349 \GTSsetSecondColor{black}%
350 \GTSincidence/%
351 \GTSsetMainColor{\GTS@mainColorBackup}%
352 \GTSsetSecondColor{\GTS@secondColorBackup}%
353 }
354
355 \GTS@newcommand{\GTSiM}{\GTSincidenceMonochrome/}
```

`\GTSincidenceMonochromeRelation` This macro will draw a monochrome incidence symbol with relation spacing in math mode.

```

Relation 356 \GTS@newcommand{\GTSincidenceMonochromeRelation}{%
\GTSiMR 357 \mathrel{\GTSincidenceMonochrome/}%
358 }
359 \GTS@newcommand{\GTSiMR}{\GTSincidenceMonochromeRelation/}
```

`\GTSnegatedIncidence-` This macro will draw a monochrome negated incidence symbol.

```

Monochrome 360 \GTS@newcommand{\GTSnegatedIncidenceMonochrome}{%
\GTSnIM 361 \def\GTS@drawSlash{1}%
362 \GTSincidenceMonochrome/%
363 \def\GTS@drawSlash{}%
364 }
365
366 \GTS@newcommand{\GTSnIM}{\GTSnegatedIncidenceMonochrome/}

```

`\GTSnegatedIncidence-` This macro will draw a monochrome negated incidence symbol with relation spacing in math mode.

```

MonochromeRelation
\GTSnIMR 367 \GTS@newcommand{\GTSnegatedIncidenceMonochromeRelation}{%
368 \mathrel{\GTSnegatedIncidenceMonochrome/}%
369 }
370 \GTS@newcommand{\GTSnIMR}{\GTSnegatedIncidenceMonochromeRelation/}

```

`\GTSoutIncidenceMonochrome` This macro will draw a monochrome directed incidence upward (out(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

```

\GTSoIM 371 \GTS@newcommand{\GTSoutIncidenceMonochrome}{%
372 \def\GTS@drawUpArrow{1}%
373 \GTSincidenceMonochrome/%
374 \def\GTS@drawUpArrow{}%
375 }
376
377 \GTS@newcommand{\GTSoIM}{\GTSoutIncidenceMonochrome/}

```

`\GTSoutIncidenceMonochrome-` This macro will draw a monochrome directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

```

Relation
\GTSoIMR 378 \GTS@newcommand{\GTSoutIncidenceMonochromeRelation}{%
379 \mathrel{\GTSoutIncidenceMonochrome/}%
380 }
381 \GTS@newcommand{\GTSoIMR}{\GTSoutIncidenceMonochromeRelation/}

```

`\GTSnegatedOutIncidence-` This macro will draw a monochrome negated directed incidence upward (out(ward) incidence) symbol.

```

Monochrome
\GTSnOIM 382 \GTS@newcommand{\GTSnegatedOutIncidenceMonochrome}{%
383 \def\GTS@drawSlash{1}%
384 \GTSoutIncidenceMonochrome/%
385 \def\GTS@drawSlash{}%
386 }
387
388 \GTS@newcommand{\GTSnOIM}{\GTSnegatedOutIncidenceMonochrome/}

```

`\GTSnegatedOutIncidence-` This macro will draw a monochrome negated directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

```

MonochromeRelation
\GTSnOIMR 389 \GTS@newcommand{\GTSnegatedOutIncidenceMonochromeRelation}{%
390 \mathrel{\GTSnegatedOutIncidenceMonochrome/}%
391 }
392 \GTS@newcommand{\GTSnOIMR}{\GTSnegatedOutIncidenceMonochromeRelation/}

```

`\GTSinIncidenceMonochrome` This macro will draw a monochrome directed incidence downward (in(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

```

\GTSiIM 393 \GTS@newcommand{\GTSinIncidenceMonochrome}{%
394 \def\GTS@drawDownArrow{1}%
395 \GTSincidenceMonochrome/%
396 \def\GTS@drawDownArrow{}%
397 }
398
399 \GTS@newcommand{\GTSiIM}{\GTSinIncidenceMonochrome/}

```

`\GTSinIncidenceMonochrome-` This macro will draw a monochrome directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

```

Relation
\GTSiIMR

```



```

400 \GTS@newcommand{\GTSinIncidenceMonochromeRelation}{%
401 \mathrel{\GTSinIncidenceMonochrome/}%
402 }
403 \GTS@newcommand{\GTSiIMR}{\GTSinIncidenceMonochromeRelation/}

\GTSnegatedInIncidence- This macro will draw a monochrome negated directed incidence downward (in(ward) inci-
Monochrome dence) symbol.
\GTSnIIM 404 \GTS@newcommand{\GTSnegatedInIncidenceMonochrome}{%
405 \def\GTS@drawSlash{1}%
406 \GTSinIncidenceMonochrome/%
407 \def\GTS@drawSlash{}%
408 }
409
410 \GTS@newcommand{\GTSnIIMR}{\GTSnegatedInIncidenceMonochrome/}

\GTSnegatedInIncidence- This macro will draw a monochrome negated directed incidence downward (in(ward) inci-
MonochromeRelation dence) symbol with relation spacing in math mode.
\GTSnIIMR 411 \GTS@newcommand{\GTSnegatedInIncidenceMonochromeRelation}{%
412 \mathrel{\GTSnegatedInIncidenceMonochrome/}%
413 }
414 \GTS@newcommand{\GTSnIIMR}{\GTSnegatedInIncidenceMonochromeRelation/}

415 \endinput
416 \</package>

```

4 Acknowledgments

Thank You God! Thank You Father! Thank You Jesus! Thank You Holy-Spirit!

5 Change History

v0.1.0 – 2026-08-01	v1.1.1 – 2026-08-30
General: Initial version 1	General: Small corrections and improvements 1
v1.0.0 – 2026-08-26	
General: First public release 1	v1.2.0 – 2026-09-05
v1.1.0 – 2026-08-29	General: CTAN URL, new macros for same set of symbols 1
General: Added a memo 1	

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